



Indian Institute of Technology Guwahati

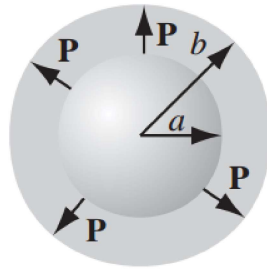
Department of Physics

PH102

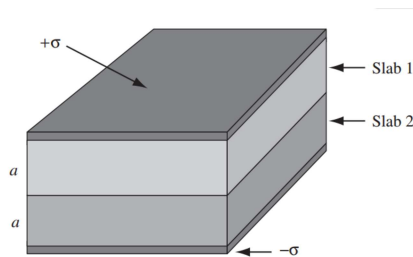
Tutorial-6

Date: 20-02-2024

1. A thick spherical shell (inner radius a , outer radius b) is made of dielectric material with a “frozen-in” polarization $\vec{P}(\vec{r}) = \frac{k}{r}\hat{r}$, where k is a constant and r is the distance from the centre. Find $\vec{D}$ and $\vec{E}$ in all three regions. Note that there is no free charge in the problem.

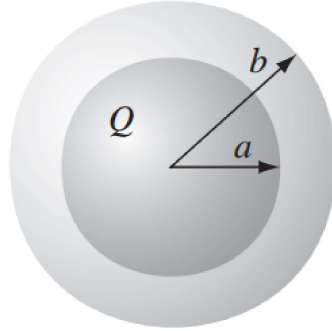


2. The space between the plates of a parallel-plate capacitor is filled with two slabs of linear dielectric material. Each slab has thickness a , so the total distance between the plates is $2a$. Slab 1 has a dielectric constant of 2, and slab 2 has a dielectric constant of 1.5. The free charge density on the top plate is σ and on the bottom plate $-\sigma$.



- Find the electric displacement $\vec{D}$ in each slab.
- Find the electric field $\vec{E}$ in each slab.
- Find the polarization $\vec{P}$ in each slab.
- Find the potential difference between the plates.
- Find the location and amount of all bound charge.

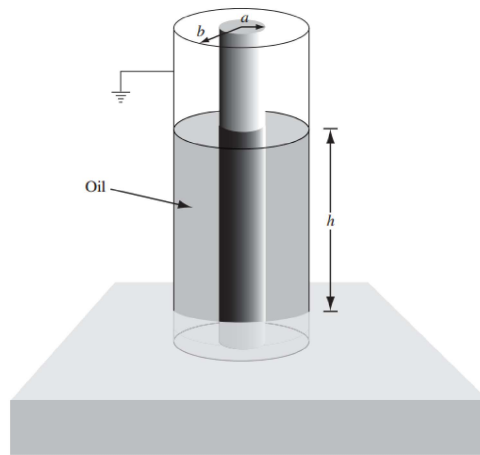
3. A spherical conductor of radius a carries a charge Q . It is surrounded by a linear dielectric material of susceptibility χ_e , out to radius b . Find the energy of this configuration.



4. Two long coaxial cylindrical metal tubes (inner radius a , outer radius b), stand vertically in a tank of dielectric oil (susceptibility χ_e , mass density ρ). The inner one is maintained at potential V , and the outer one is grounded. To what height h does the oil rise in the space between the tubes? Assume that the force experienced by the oil column is given by

$$F = \frac{1}{2} V^2 \frac{dC}{dh}$$

where C is the capacitance of the given system.



5. Two homogeneous dielectric regions 1 ($s \leq 4\text{cm}$) and 2 ($s \geq 4\text{cm}$) have dielectric constants 3.5 and 1.5, respectively. If $\vec{D}_2 = 12\hat{s} - 6\hat{\phi} + 9\hat{z}$; calculate
- $\vec{E}_1$ and $\vec{D}_1$,
 - $\vec{P}_2$ and ρ_2 ,
 - the energy density for each region.

Sol 1. Given that,

$$\vec{P}(\vec{r}) = \begin{cases} 0; & r < a \\ \frac{k}{r} \hat{r}; & a < r < b \\ 0; & r > b \end{cases} \quad \text{--- (1)}$$

("frozen-in" polarization)

and $Q_f = 0$ ($\sigma_f = 0$) --- (2)

Now, $\oint \vec{D} \cdot d\vec{a} = Q_{\text{enc}}$ --- (3)

Since $Q_f = 0 \therefore \vec{D} = 0$ --- (4)

for $r < a$; $a < r < b$; $r > b$

Now, $\vec{D} = \epsilon_0 \vec{E} + \vec{P}$ --- (5)

and using (4)

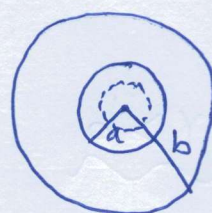
$$\vec{E} = -\frac{\vec{P}}{\epsilon_0} \quad \text{--- (6)}$$

$\therefore$ using (1) in (6),

$$\vec{E}(\vec{r}) = \begin{cases} 0; & r < a \\ -\frac{k}{\epsilon_0 r} \hat{r}; & a < r < b \\ 0; & r > b \end{cases} \quad \text{--- (7)}$$

Alternate solution

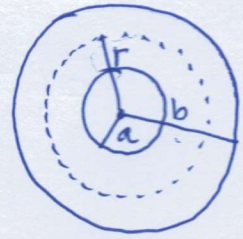
For $r < a$: $Q_f = 0$ & no other charge present



$\therefore Q_{\text{enc.}} = 0 \therefore \vec{E} = 0; \vec{D} = 0$ --- (1a)

For $a < r < b$ $\vec{D} = 0$ ($\because Q_f = 0$) — (1b)

$$\oint \vec{E} \cdot d\vec{A} = Q_{enc.}/\epsilon_0$$



Now, the surface charge is

$$\sigma_B = \vec{P} \cdot \hat{n} = \frac{k}{r} \hat{r} \cdot (-\hat{r}) = -\frac{k}{r} \quad \text{--- (2a)}$$

$$\therefore \text{at } r = a, \sigma_B = -\frac{k}{a} \quad \text{--- (2c)}$$

$$\rho_B = -\vec{\nabla} \cdot \vec{P} = -\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{k}{r} \right) = -\frac{k}{r^2} \quad \text{--- (2d)}$$

$$\therefore Q_B = \int \sigma_B d\vec{A} = - \int_0^{2\pi} \int_0^\pi \frac{k}{a} \cdot a^2 \sin\theta d\theta d\phi$$

$$= -4\pi ka. \quad \text{--- (2e)}$$

$$Q_P = \int \rho_B dz = - \int_a^r \int_0^{2\pi} \int_0^\pi -\frac{k}{r^2} r^2 \sin\theta dr d\theta d\phi$$

$$= -4\pi k \int_a^r dr = -4\pi k(r-a) \quad \text{--- (2f)}$$

$$\therefore Q_{enc.} = Q_B + Q_P = -4\pi ka - 4\pi k(r-a)$$

$$= -4\pi kr \quad \text{--- (2g)}$$

$$\therefore E(4\pi r^2) = -\frac{4\pi kr}{\epsilon_0} \Rightarrow \vec{E} = -\frac{k}{\epsilon_0 r} \hat{r} \quad \text{--- (2h)}$$

For $r > b$: $\vec{D} = 0$ ($\because Q_f = 0$) — (3a)

Now,

$$\begin{aligned} Q_B &= \int \sigma_B dA \Big|_{r=a} + \int \sigma_B dA \Big|_{r=b}^* \\ &= -\frac{k}{a} \int_0^\pi \int_0^{2\pi} a^2 \sin\theta d\theta d\phi + \frac{k}{b} \int_0^\pi \int_0^{2\pi} b^2 \sin\theta d\theta d\phi \\ &= -4\pi ka + 4\pi kb \quad \text{--- (3b)} \end{aligned}$$

and

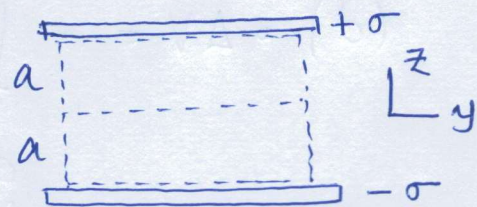
$$\begin{aligned} Q_P &= \iiint \rho_B dz \\ &= -k \int_0^{2\pi} \int_0^\pi \int_a^b \frac{1}{r^2} r^2 \sin\theta dr d\theta d\phi \\ &= -4\pi k \int_a^b dr = -4\pi k(b-a) \quad \text{--- (3c)} \end{aligned}$$

$$\therefore Q_{enc.} = Q_P + Q_B = 0$$

$$\therefore E(4\pi r^2) = 0 \Rightarrow \vec{E} = 0 \quad \text{--- (3d)}$$

$$* \quad \sigma_B = \vec{P} \cdot \hat{n} = \frac{k}{r} \hat{r} \cdot (\hat{r}) = \frac{k}{r} = \frac{k}{b} \text{ at } r=b.$$

Sol. 2 The free charge on the capacitor's plates are $\pm\sigma$.



(a) Applying the boundary condition

$$\vec{D}_{\text{above}}^{\perp} - \vec{D}_{\text{below}}^{\perp} = \sigma_f \hat{n} \quad (\hat{n} \equiv \hat{z}) \quad \text{--- (1)}$$

at both the capacitors, we get-

$$D = \sigma$$

$$\text{or } \vec{D} = -\sigma \hat{z}$$

--- (2)*

and is the same in each of the two slabs since it depends only on the free charge.

$$(b) \vec{D} = \epsilon \vec{E} = \epsilon_r \epsilon_0 \vec{E} \quad \text{--- (3)}$$

$$\therefore \vec{E}_1 = \frac{\vec{D}}{\epsilon_r \epsilon_0} = -\frac{\sigma}{2\epsilon_0} \hat{z} \quad \text{--- (4)}$$

$$\& \vec{E}_2 = \frac{\vec{D}}{\epsilon_r \epsilon_0} = -\frac{2\sigma}{3\epsilon_0} \hat{z} \quad \text{--- (5)}$$

$$(c) \vec{P} = \epsilon_0 \chi_e \vec{E} = \epsilon_0 (\epsilon_r - 1) \vec{E} \quad \text{--- (6)}$$

$$\therefore \vec{P}_1 = \epsilon_0 (2-1) \left(-\frac{\sigma}{2\epsilon_0} \hat{z} \right) = -\frac{\sigma}{2} \hat{z} \quad \text{--- (7)}$$

(using (4))

$$\text{and } \vec{P}_2 = \epsilon_0 (1.5-1) \left(-\frac{2\sigma}{3\epsilon_0} \hat{z} \right) = -\frac{\sigma}{3} \hat{z} \quad \text{--- (8)}$$

(using (5))

* Gauss's law $\oint \vec{D} \cdot d\vec{A} = Q_f$ gives the similar result.

$$(d) \Delta V = - \int_{ref}^r \vec{E} \cdot d\vec{l} = - \int_0^{2a} \vec{E} \cdot d\vec{l} \quad (d\vec{l} = \hat{z} dl) \quad \text{--- (9)}$$

$$= - \int_0^a \vec{E}_1 \cdot d\vec{l}_1 - \int_a^{2a} \vec{E}_2 \cdot d\vec{l}_2$$

$$= E_1 a + E_2 a = \frac{\sigma a}{2\epsilon_0} + \frac{2\sigma a}{3\epsilon_0}$$

$$\text{or } \Delta V = \frac{7\sigma a}{6\epsilon_0} \quad \text{--- (10)}$$

$$(e) \sigma_b = \vec{P} \cdot \hat{n} \quad \text{--- (11)}$$

At the top of slab 1,

$$\sigma_{b_1} = \vec{P}_1 \cdot \hat{z} = -\frac{\sigma}{2} \quad \text{--- (12)}$$

At the bottom of slab 1,

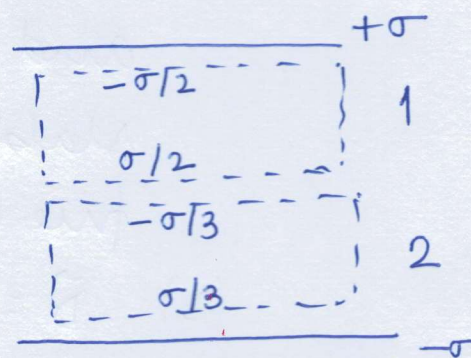
$$\sigma_{b_1} = \vec{P}_1 \cdot (-\hat{z}) = \frac{\sigma}{2} \quad \text{--- (13) (using (7))}$$

At the top of slab 2,

$$\sigma_{b_2} = \vec{P}_2 \cdot \hat{z} = -\frac{\sigma}{3} \quad \text{--- (14)}$$

and at the bottom of slab 2,

$$\sigma_{b_2} = \vec{P}_2 \cdot (-\hat{z}) = \frac{\sigma}{3} \quad \text{--- (15) (using (8))}$$

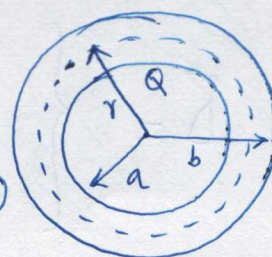


Q3

The energy of the configuration can be calculated using the relation

$$W = \frac{1}{2} \int_{\text{all space}} \vec{D} \cdot \vec{E} d\tau \quad \text{--- (1)}$$

for $r < a$, $\vec{E} = 0$, $\vec{D} = 0$ --- (2)



(Sphere is a conductor & hence no electric field and also, no free charge)

for $a < r < b$, consider a Gaussian sphere of radius r

$$\oint \vec{D} \cdot d\vec{a} = Q_{\text{enc}} = Q \quad (\text{given})$$

$$\text{or } \int_0^\pi \int_0^{2\pi} D \hat{r} \cdot \hat{r} r^2 \sin\theta d\theta d\phi = Q$$

$\theta=0 \quad \phi=0$

$$\text{or, } \vec{D} = \frac{Q}{4\pi r^2} \hat{r} \quad \text{--- (3)}$$

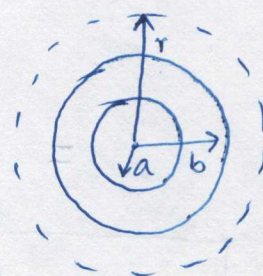
$$\therefore \vec{D} = \epsilon \vec{E} = \epsilon_0(1+\chi_e) \vec{E}$$

$$\Rightarrow \vec{E} = \frac{Q}{4\pi\epsilon_0(1+\chi_e)r^2} \hat{r} \quad \text{--- (4)}$$

for $r > b$

$$\vec{D} = \frac{Q}{4\pi r^2} \hat{r} \quad \text{--- (5)}$$

(same as (3)).



and $\vec{D} = \epsilon_0 \vec{E}$

$$\Rightarrow \vec{E} = \frac{Q}{4\pi\epsilon_0 r^2} \hat{r} \quad \text{--- (6)}$$

$\therefore$ We have

$$\vec{D} = \begin{cases} 0, & r < a \\ \frac{Q}{4\pi r^2} \hat{r}, & r > a \end{cases}$$

and

$$\vec{E} = \begin{cases} 0, & r < a \\ \frac{Q}{4\pi\epsilon_0(1+\chi_e)r^2} \hat{r}, & a < r < b \\ \frac{Q}{4\pi\epsilon_0 r^2} \hat{r}, & r > b \end{cases}$$

--- (7)

$$\begin{aligned} \therefore W &= \frac{1}{2} \left[\int_0^\infty \vec{D} \cdot \vec{E} d\tau \right] \\ &= \frac{1}{2} \left[\int_0^a \vec{D} \cdot \vec{E} d\tau + \int_a^b \vec{D} \cdot \vec{E} d\tau + \int_b^\infty \vec{D} \cdot \vec{E} d\tau \right] \\ &= \frac{1}{2} \left[(0) + \left(\frac{Q}{4\pi} \right)^2 \frac{1}{\epsilon_0(1+\chi_e)} \int_a^b \frac{1}{r^4} r^2 dr \int_0^\pi \sin\theta d\theta \int_0^{2\pi} d\phi \right. \\ &\quad \left. + \left(\frac{Q}{4\pi} \right)^2 \frac{1}{\epsilon_0} \int_b^\infty \frac{1}{r^4} r^2 dr \int_0^\pi \sin\theta d\theta \int_0^{2\pi} d\phi \right] \\ &= \frac{1}{2} \left(\frac{Q}{4\pi} \right)^2 \cdot \frac{4\pi}{\epsilon_0} \left[\frac{1}{1+\chi_e} \left(\frac{1}{a} - \frac{1}{b} \right) + \frac{1}{b} \right] \end{aligned}$$

or $W = \frac{Q^2}{8\pi\epsilon_0} \left\{ \frac{1}{a} + \frac{\chi_e}{b} \right\} \left(\frac{1}{1+\chi_e} \right) \quad \text{--- (8)}$

Sol. 24

The configuration can be thought of two capacitors in parallel (with and without any dielectric material, say).

Let λ and λ' be the linear charge densities of the cable in air and in the oil. Let L be the length of the cylindrical cable.

For the cable in air

$$\vec{E} = \frac{\lambda}{2\pi\epsilon_0 r} \hat{r} \quad \text{for } a < r < b \quad \text{--- (1)}$$

and for the cable in oil

$$\vec{E}' = \frac{\lambda'}{2\pi\epsilon r} \hat{r} \quad \text{for } a < r < b \quad \text{--- (2)}$$

The potential difference between the cables in air is

$$V_a = - \int_b^a \vec{E} \cdot d\vec{l} = - \frac{\lambda}{2\pi\epsilon_0} \int_b^a \frac{1}{r} dr = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{b}{a}\right) \quad \text{--- (3)}$$

Similarly, in the oil

$$V_o = - \int_b^a \vec{E}' \cdot d\vec{l} = - \frac{\lambda'}{2\pi\epsilon} \int_b^a \frac{1}{r} dr = \frac{\lambda'}{2\pi\epsilon} \ln\left(\frac{b}{a}\right) \quad \text{--- (4)}$$

Since the two are in parallel

$$\therefore V_o = V_a$$

$$\Rightarrow \frac{\lambda}{\epsilon_0} = \frac{\lambda'}{\epsilon} \quad \text{--- (5)}$$

Now, the total charge

$$\begin{aligned} Q &= \lambda(L-h) + \lambda'h \\ &= \lambda(L-h) + \frac{\lambda\epsilon}{\epsilon_0} h \\ &= \lambda(L-h) + \lambda(1+\chi_e)h, \quad \epsilon = \epsilon_0(1+\chi_e) \end{aligned}$$

$$\text{or } Q = \lambda(L + \chi_e h) \quad \text{--- (6)}$$

$\therefore$ The capacitance is

$$\begin{aligned} C &= \frac{Q}{V} = \frac{\lambda(L + \chi_e h)}{\lambda \ln(b/a)} (2\pi\epsilon_0) \\ &= 2\pi\epsilon_0 \frac{(L + \chi_e h)}{\ln(b/a)} \end{aligned}$$

$\therefore$ Force experienced by the oil

$$F_o = \frac{1}{2} V^2 \frac{dC}{dh} = \frac{\pi\epsilon_0\chi_e V^2}{\ln(b/a)} \quad \text{--- (7)}$$

The oil column experiences the gravitational force

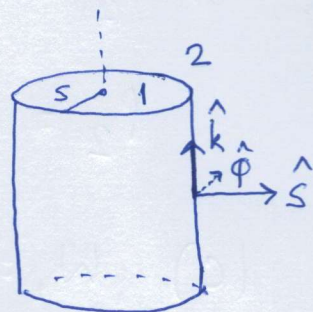
$$F = \rho\pi(b^2 - a^2)gh \quad \text{--- (8)}$$

$\therefore$ When ~~these~~ these forces balance each other the height is then

$$\boxed{h = \frac{\epsilon_0\chi_e V^2}{\rho(b^2 - a^2)g \ln(b/a)}} \quad \text{--- (9)}$$

Sol.5 Normal to the surface is $\hat{s}$ direction.
Parallel to the surface includes $\hat{\phi}$ & $\hat{k}$.

$$\left. \begin{aligned} \vec{D}_2 &= 12\hat{s} - 6\hat{\phi} + 9\hat{z} \\ \epsilon_{r_1} &= 3.5 \quad \epsilon_{r_2} = 1.5 \end{aligned} \right\} \text{--- (1)}$$



(a) There is no free charge.

$\therefore$ the normal component of $\vec{D}$ is continuous,

$$\vec{D}_1^\perp = \vec{D}_2^\perp = 12\hat{s} \text{--- (2)}$$

The parallel component of $\vec{E}$ is also continuous,

$$\therefore \vec{E}_1^\parallel = \vec{E}_2^\parallel \Rightarrow \frac{\vec{D}_1^\parallel}{\epsilon_1} = \frac{\vec{D}_2^\parallel}{\epsilon_2} \quad (\epsilon = \epsilon_r \epsilon_0) \text{--- (3)}$$

$$\text{Now, } \vec{D}_2^\parallel = -6\hat{\phi} + 9\hat{z} \text{--- (4)}$$

$$\therefore \vec{D}_1^\parallel = \frac{\epsilon_1}{\epsilon_2} \vec{D}_2^\parallel = \frac{3.5}{1.5} (-6\hat{\phi} + 9\hat{z})$$

$$\text{or } \vec{D}_1^\parallel = -14\hat{\phi} + 21\hat{z} \text{--- (5)}$$

$$\therefore \vec{D}_1 = \vec{D}_1^\parallel + \vec{D}_1^\perp = 12\hat{s} - 14\hat{\phi} + 21\hat{z} \text{--- (6)}$$

$$\text{Also, } \vec{E}_1 = \frac{\vec{D}_1}{\epsilon_1} = \frac{12\hat{s} - 14\hat{\phi} + 21\hat{z}}{3.5\epsilon_0}$$

$$\text{or, } \vec{E}_1 = \frac{1}{\epsilon_0} \left(\frac{24}{7}\hat{s} - 4\hat{\phi} + 6\hat{z} \right) \text{--- (7)}$$

$$(b) \vec{P}_2 = \epsilon_0 \chi_{e_2} \vec{E}_2 = \epsilon_0 (\epsilon_{r_2} - 1) \frac{\vec{D}_2}{\epsilon_{r_2} \epsilon_0} \text{--- (8)}$$

$$= \frac{0.5}{1.5} (12\hat{s} - 6\hat{\phi} + 9\hat{z})$$

$$\text{or } \vec{P}_2 = 4\hat{s} - 2\hat{\phi} + 3\hat{z} \quad \text{--- (9)}$$

$$P_2 = -\vec{\nabla} \cdot \vec{P}_2 = -\frac{1}{s} \frac{\partial}{\partial s} (4s) = -\frac{4}{s} \quad \text{--- (10)}$$

$$(c) \quad W_1 = \frac{1}{2} \vec{D}_1 \cdot \vec{E}_1 \quad \text{--- (11)}$$

$$= \frac{1}{2\epsilon_0} (12\hat{s} - 14\hat{\phi} + 21\hat{z}) \cdot \left(\frac{24}{7}\hat{s} - 4\hat{\phi} + 6\hat{z} \right) \quad \text{(using (6) \& (7))}$$

$$= \left(\frac{144}{7} + 28 + 63 \right) \frac{1}{\epsilon_0}$$

$$= \frac{781}{7\epsilon_0} \quad \text{--- (12)}$$

$$W_2 = \frac{1}{2} \vec{D}_2 \cdot \vec{E}_2 \quad \text{--- (13)}$$

$$= \frac{1}{2} \vec{D}_2 \cdot \frac{\vec{D}_2}{\epsilon_{r_2} \epsilon_0} = \frac{1}{3\epsilon_0} (144 + 36 + 81)$$

$$= \frac{261}{3\epsilon_0} = \frac{87}{\epsilon_0} \quad \text{(using (1))}$$

$$\text{--- (14)}$$